Module 12: Windows Server Basics & Installation

1. What two options are provided in the type of installation window during Windows Server 2016 installation?

- **There are two installation options:**

1. Server Core

- Minimal installation (no GUI)
- Uses command line / PowerShell
- Less resource usage
- More secure (smaller attack surface)
- Best for production environments

2. Server with Desktop Experience

- Full graphical user interface (GUI)
- Easy to use and manage
- Requires more RAM and storage
- Suitable for beginners and administrative tasks

2. Write the step How to configure server step by step?

After installation:

1. Set Administrator password
2. Rename the server
3. Configure IP address (static recommended)
4. Set DNS settings

5. Join domain or workgroup
6. Install roles and features (via Server Manager)
7. Enable Remote Desktop (optional)
8. Run Windows Updates
9. Configure firewall and security settings

3. What are the Pre installation tasks?

Before installing Windows Server:

- Check hardware compatibility
- Verify minimum requirements
- Backup important data
- Prepare bootable USB/DVD
- Decide partition structure
- Collect license key
- Plan server roles (AD, DNS, etc.)

4. What are the Post installation tasks?

After installation:

- Install latest updates
- Configure network settings
- Rename server
- Activate Windows
- Install server roles
- Configure Active Directory (if needed)
- Setup backup system
- Enable security policies

5. What is the standard upgrade path for Windows Server?

Supported upgrade path:

- Windows Server 2012 → 2016
- Windows Server 2012 R2 → 2016

Direct upgrade from very old versions (like 2008) is not recommended

6. What is the Physical structure of AD?

Includes:

- Domain Controllers (DCs) – store AD database
- Sites – represent physical locations
- Subnets – IP network segments

Used for replication and network optimization

7. What is the Logical components of Active Directory?

Includes:

- Forest (top level)
- Tree
- Domain
- Organizational Units (OUs)
- Objects (users, computers, groups)

8. What is the Full form Of LDAP?

Lightweight Directory Access Protocol

Used to access and manage directory services like Active Directory

9. What is the location of the AD database?

- File: NTDS.dit
- Location:
- C:\Windows\NTDS\

10. What is child DC?

- A domain controller in a child domain
- Child domain is part of a parent domain
- Shares trust relationship with parent domain

11. Explain the term forest in AD

- Highest level of AD structure
- Contains multiple domains
- Shares:
    - Schema
    - Configuration
    - Global Catalog

Example: company.com → sales.company.com → hr.company.com

12. What is Active Directory? Check all that apply.

● **An open-source directory server**

● **A Windows-only implementation of a directory server**

● **Microsoft's implementation of a directory server**

● **An LDAP-compatible directory server**

✓ Microsoft’s implementation of a directory server
✓ LDAP-compatible directory server

13. When you create an Active Directory domain, what's the name of the default user account?

● Superuser

● Root

● Username

● Administrator

✓ Administrator

14. AD domain provides which of the following advantages? Check all that apply.

● Centralized authentication

● More detailed logging

● Centralized management with GPOs

● Better performance

✓ Centralized authentication
✓ More detailed logging
✓ Centralized management using GPO

15. What are the minimum hardware requirements for installing Windows Server 2016?CPU: 1.4 GHz 64-bit

- RAM: 512 MB (2 GB recommended)
- Storage: 32 GB minimum
- Network: Gigabit adapter

16. Explain the different editions of Windows Server 2016 and their features.

1. Essentials
    - For small businesses
    - Max 25 users
2. Standard
    - Basic features
    - Limited virtualization (2 VMs)
3. Datacenter
    - Full features
    - Unlimited virtualization
    - Advanced storage & networking

17. Walk through the steps of installing Windows Server 2016 using GUI mode.

1. Boot from USB/DVD
2. Select language
3. Click “Install Now”
4. Choose edition
5. Accept license
6. Select Custom Installation
7. Choose disk/partition
8. Install OS
9. Restart system

18. Describe the steps for installing Windows Server 2016 in Server Core mode.

Same as GUI installation, but:

- Select Server Core edition
- After installation → manage using:
    - PowerShell
    - CMD
    - Remote tools

19. How do you configure network settings during Windows Server 2016 installation? After install:

- Open Server Manager
- Go to Local Server
- Set:
    - Static IP
    - Subnet mask
    - Default gateway
    - DNS server

20. Explain the process of promoting a Windows Server to a domain controller.

Steps:

1. Install AD DS role
2. Click Promote this server
3. Choose:
    - New forest OR existing domain
4. Set domain name
5. Configure DNS
6. Set DSRM password

7. Install & restart

21. What is Active Directory Domain Services (AD DS), and what are its key components?

Active Directory Domain Services

Provides:

- Authentication
- Authorization
- Directory services

Components:

- Domains
- Trees
- Forests
- Domain Controllers
- OUs

22. How do you create a new Active Directory user account in Windows Server ?

Steps:

1. Open Active Directory Users and Computers
2. Right-click OU
3. Click New → User
4. Enter details
5. Set password
6. Finish

23. Explain the process of creating and managing Group Policy Objects (GPOs) in Windows Server 2016 or 2019.

Used to control:

- Security settings
- Software installation
- User restrictions

Steps:

1. Open Group Policy Management
2. Create new GPO
3. Edit settings
4. Link to domain/OU

24. What are Organizational Units (OUs) in Active Directory, and how do you use them?

- Containers inside a domain
- Used to organize users, groups, computers
- Helps apply GPOs easily

Example:

- HR OU
- IT OU
- Sales OU

25. Describe the process of delegating administrative privileges in Active Directory.

Steps:

1. Open AD Users & Computers

2. Right-click OU
3. Select Delegate Control
4. Choose user/group
5. Assign permissions

Used to give limited admin rights without full control